



Each press hits `POST /api/device/{station_id}/event` with `type="pia.button"` + `payload={action, by}`. The Code Blue button gates on a `confirm()` to prevent fat-finger fires.

On press, the device: 1. POSTs the event. 2. Plays a one-shot WAV (from the M49 clinical-alarms library). 3. Flashes the entire screen frame in an alternating colour matched to the kind (blue for `call_bell`, amber for `bed_alarm`, red for `code_blue`, green for `intercom`). 4. The footer's "last event" stamp updates so the student knows the press registered even before any audio plays.

2.3 Server-side button router — `_handle_pia_button`

In `portal/devices/routes.py` the `api_device_event` route detects PIA button events and dispatches:

action	Effect
<code>call_bell</code>	<code>alarm.injected device_event</code> with <code>tone=call_bell</code> → alarm bus picks it up at <code>severity=info</code>
<code>bed_alarm</code>	<code>alarm.injected device_event</code> with <code>tone=bed_alarm</code> → alarm bus picks it up at <code>severity=warning</code>
<code>code_blue</code>	<code>scenes.apply(code.blue)</code> — the existing M7 scene — which writes chart events + raises a room-wide code-blue alarm
<code>intercom_request</code>	<code>comm.intercom_request</code> chart event + transcript line ("■ Intercom requested")

This re-uses the existing M26/M27 plumbing — the PIA does **not** introduce a new alarm bus channel; it just emits the same kinds of events the Nursing Station's "inject alarm" buttons emit, so the existing alarm board / silence / clear flows work unchanged.

2.4 Room-wide code-blue cascade

When any PIA fires a Code Blue, the scene apply writes a `code.blue` alarm into the room's alarm bus. Every PIA in the room polls `/api/room/alarms` every 3 s (`pollCascade` in `pia_app.js`).

If ANY code-blue alarm is active anywhere in the room, every PIA:

- Shows the **■ CODE BLUE — Bed N** cascade banner across the top, reading the originating encounter's label so the bedside student knows WHERE the code is, even though THEIR bed is calm.
- Adds the `pia-cascade-active` class to the frame, which keeps the red flash continuous until the alarm clears.
- Plays the `03_code_blue.wav` once, ONLY when the active set changes (`_cascadeKey` dedupes by sorted `alarm_ids`).

This satisfies the operator ask: *"the code blue should also cause the other Patient integrated alarm state the location"*. Same mechanism notifies the Nursing Station (via the alarm bus already displayed there) and the instructor's encounter console (via the M27 alarm card).

2.5 Instructor mirror panel

When a PIA station is bound to an encounter, the encounter console's Devices card renders a four-button mirror panel inline (■ Instructor mirror – fire any event on this bed's PIA tablet). Each button POSTs the same `pia.button` event so the instructor can trigger any of the four actions without leaving the encounter view — essential for "scripted" calls in pre-planned drills, or for firing the alarm if the patient role-player isn't reachable.

3. Files touched

New files

- `portal/devices/pia/pia_v1/spec.json` — declares 4 controls + 3 alarm tones.
- `portal/devices/pia/pia_v1/skin.svg` — registry-stub SVG (PIA renders a custom template).
- `portal/templates/device_pia.html` — 4-button + cascade banner tablet UI.
- `portal/static/pia_app.js` — press handler, sound playback, screen flash, cascade poller, heartbeat.
- `portal/static/pia_app.css` — dark theme + per-kind flash keyframes.
- `tests/v7/test_patient_integrated_alarm.py` — 13 acceptance tests.
- `docs/module_guides/M51_patient_integrated_alarm.md` — this guide.

Modified

- `portal/devices/registry.py` — `patient_integrated_alarm` added to `KIND_DIRS` and `REFERENCE_MODELS`.
- `portal/devices/engine/state_machine.py` — `make_engine` branches to a thin `PiaEngine` (no custom reducer; the base no-op reducer is enough because PIA effects are side-effects routed in `routes._handle_pia_button`, not folded into device-station state).
- `portal/devices/routes.py` — `device_app` branches to `device_pia.html` for PIA kind; `api_device_event` calls `_handle_pia_button` for `pia.button` events.
- `portal/static/encounter_console.js` — ■ Patient Integrated Alarm kind label, PIA mirror-panel render in the device card, `pia` dispatch branch in `onDeviceAction`.
- `portal/static/encounter_console.css` — `.device-card-pia` + `.pia-mirror-row` + `.pia-mirror-btn` styles.

4. Acceptance — what M51 must satisfy

Source: `tests/v7/test_patient_integrated_alarm.py` (13 tests).

1. **Registry** - `patient_integrated_alarm` appears in `registry.list_kinds()`. - `registry.available_models("patient_integrated_alarm") == ["pia_v1"]`. - The spec declares `call_bell`, `bed_alarm`, `code_blue`, and `intercom` controls.
2. **Engine factory** - `make_engine(kind="patient_integrated_alarm", model="pia_v1")` returns a `PiaEngine` without raising.
3. **Device template** - GET `/device/{join_code}/{station_id}` for a PIA station serves `device_pia.html` (carries `pia-frame`, `pia-grid`, all four data-action buttons, loads `pia_app.js`). -

Does NOT serve the generic vendor-skin template.

4. **Button routing** - `call_bell` press → `alarm.injected` device alarm on `/api/room/alarms` with `kind="call_bell"`, `source="device"`, `correct_encounter_id`. - `bed_alarm` press → same shape, `kind="bed_alarm"`. - `code_blue` press from a 2-bed room → code-blue alarm visible on `/api/room/alarms` with `encounter_label` populated (so other PIAs can show *where* the code is), `severity ≥ critical`. - `intercom_request` press → `comm.intercom_request` chart event written with the `station_id`; transcript carries a "■ Intercom requested" line in the last 4 entries. - Unknown action → route still returns 200 (event persisted by engine), no alarm raised.

5. **Sound assets** - `01_bed_exit_alarm.wav`, `02_call_bell.wav`, `03_code_blue.wav` all exist on disk under `portal/static/sounds/clinical_alarms/`. - `pia_app.js` references all three by filename + carries all four `pia-flash-*` class names + cascade poller hits `/api/room/alarms`.

6. **Encounter console** - `encounter_console.js` contains the `patient_integrated_alarm` kind label. - Contains four `data-pia-action="..."` mirror buttons and a `pia.button` POST handler.

7. **CSS** - All four `@keyframes` `flash-*` rules defined plus `cascade-pulse` for the banner. - `.pia-frame.pia-flash-code-blue` and `.pia-cascade-active` rules present.

All 13 tests pass in `pytest tests/v7/test_patient_integrated_alarm.py` and the full v7 suite (387 passed, 1 skipped) shows no regression.

5. Operator demo script

1. From `/portal/control/ops` for an active multi-patient room, open any encounter. The Devices card shows the existing inline device manager. 2. Click **Add device**, pick **Patient Integrated Alarm** → `pia_v1`, give it a label (e.g. "Bed 1 PIA"). A QR code appears in the Devices card. 3. Scan the QR with a tablet — the tablet loads `device_pia.html` showing the four buttons + cascade banner area. 4. From a second encounter, repeat steps 2–3 with another tablet (or browser window simulating one). 5. **Test the four buttons** from the bedside tablet: - ■ Call Bell — screen flashes blue, soft chime plays, the Nursing Station's alarm board picks up a `call_bell` alarm scoped to this bed. - ■ Bed Alarm — screen flashes amber, bed-exit chime plays, Nursing Station picks up `bed_alarm`. - ■ Code Blue — confirm dialog appears, then screen flashes red, code-blue tone plays, **every other PIA in the room also flashes red and shows "CODE BLUE — Bed N"**. The Nursing Station's code-blue alarm + the encounter's alarm card both fire. - ■ Intercom — screen flashes green, the transcript on the encounter console shows "■ Intercom requested", and the Nursing Station's intercom UI surfaces the request. 6. From the instructor's encounter console, use the ■ **Instructor mirror** buttons in the Devices card to fire any of the four actions without touching the tablet — useful for scripted drills.

6. Things the PIA deliberately does NOT do

- **No microphone capture or 2-way audio.** Intercom uses the

existing M28 nurse→bedside one-way voice channel; the PIA just signals "patient wants the intercom open" via a chart event. The Nursing Station student or instructor opens the audio leg.

- **No alarm silence / clear from the device side.** Those are

operator + nurse-station actions. The PIA is a *patient-facing* device — silencing your own emergency alarm would defeat the point.

- **No fire-alarm button.** The M29 stub still exists for the

fire-alarm case (it's a building-level event, not a per-bed device). PIA stays focused on bedside-only actions.

- **No persistent screen lock or kiosk-mode enforcement.** Kiosk

mode is a deployment concern (iOS Guided Access, Android pin- to-app) — the device-side code just hides browser chrome via the meta tags `apple-mobile-web-app-capable` + `theme-color`.

7. Future hooks

- **PIA assignment to a Student.role.** Right now a PIA is bound

to an encounter, not a student. A future cleanup could let the patient role-player register against the PIA so transcript entries carry their `student_id`.

- **Customizable alarm tones per encounter.** The 3 tones are

hard-coded to M49 WAVs. Curriculum authors could pick alternate tones in the wizard.

- **Patient-side accept/decline of the intercom open.** Right now

the request fires once; the Nursing Station controls when to open audio. A future module could add a "Nurse is opening intercom" notification on the PIA with an accept/decline affordance.

Closes: the operator's M51 ask in full. No follow-on tickets identified.